

Math 372: Homework 03

Due Friday, September 11

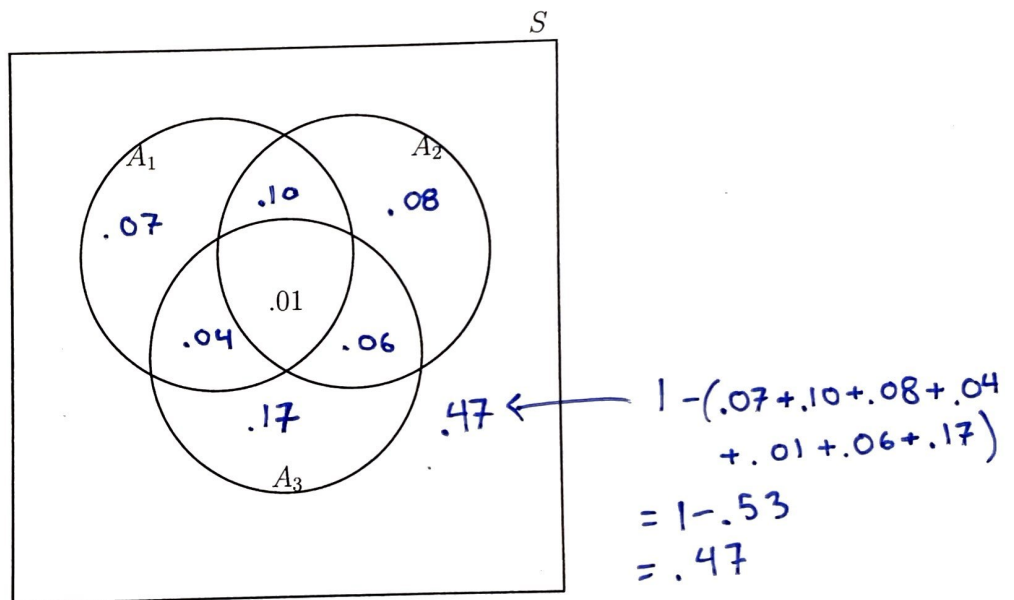
Problem 1 (Sets). A consulting firm has bids on 3 projects. Let

$$A_i = [\text{awarded project } i], \quad i \in \{1, 2, 3\}$$

Suppose that

$$\begin{aligned} \mathbb{P}[A_1 \cap A_2 \cap A_3] &= .01 \\ \mathbb{P}[A_1] &= .22, \quad \mathbb{P}[A_2] = .25, \quad \mathbb{P}[A_3] = .28 \\ \mathbb{P}[A_1 \cap A_2] &= .11, \quad \mathbb{P}[A_1 \cap A_3] = .05, \quad \mathbb{P}[A_2 \cap A_3] = .07 \end{aligned}$$

- (a) In the following Venn diagram, the three overlapping circles represent events A_1, A_2 , and A_3 , all contained within the sample space box S . Each enclosed region represents an event, all of which have some probability of occurring. I've already filled in the probability of one of the events (namely, the event $A_1 \cap A_2 \cap A_3$). Using the probabilities above, fill in the probabilities of other 7 regions:



- (b) Express in words each of the following events, and compute the probability of each event:

- (i.) $A_1 \cup A_2$ $P[A_1 \cup A_2] = .07 + .10 + .08 + .04 + .01 + .06 = .36$
- (ii.) $A_1^c \cap A_2^c = (A_1 \cup A_2)^c$ $P[(A_1 \cup A_2)^c] = 1 - P[A_1 \cup A_2] = 1 - .36 = .64$
- (iii.) $A_1 \cup A_2 \cup A_3$ $P[A_1 \cup A_2 \cup A_3] = .53$
- (iv.) $A_1^c \cap A_2^c \cap A_3^c$ "didn't get any contract" Prob = .47
- (v.) $A_1^c \cap A_2^c \cap A_3$ "got project 3 but not 1 or 2" Prob = .17
- (vi.) $(A_1^c \cap A_2^c) \cup A_3$ "either got project 3, or didn't get 1 and 2"
 Prob = .28 + .47 = .75